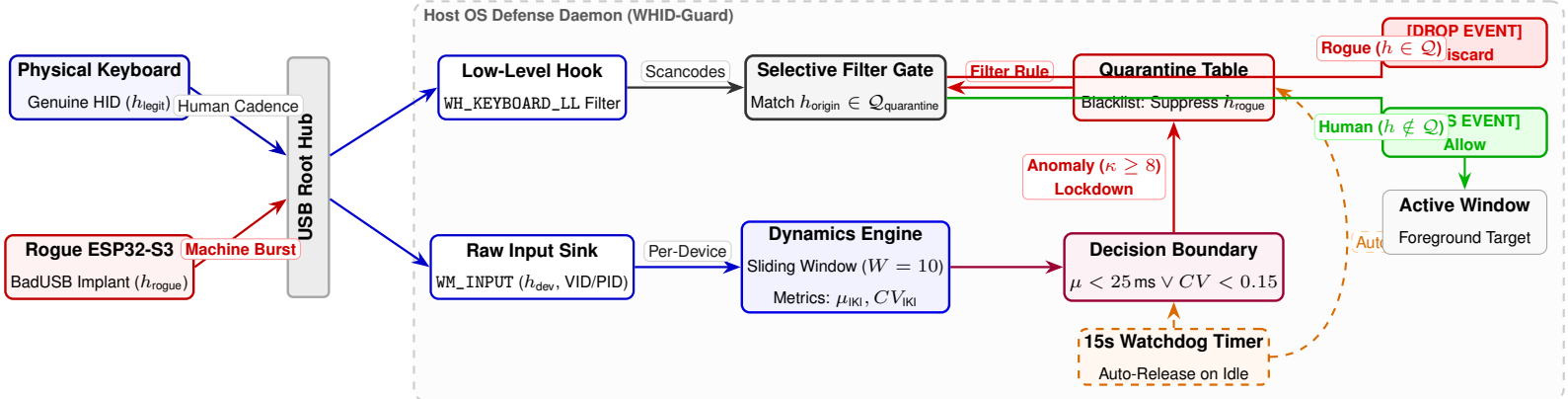

Host OS Defense Daemon (WHID-Guard)
Physical Keyboard
Genuine HID (h_{legit})
Human Cadence
Rogue ESP32-S3
BadUSB Implant (h_{rogue})
Machine Burst
USB Root Hub
Low-Level Hook
WH_KEYBOARD_LL Filter
Scancodes
Raw Input Sink
WM_INPUT (h_{dev}, VID/PID)
Per-Device
Selective Filter Gate
Match $h_{\text{origin}} \in \mathcal{Q}_{\text{quarantine}}$
Filter Rule
Dynamics Engine
Sliding Window ($W = 10$)
Metrics: $\mu_{\text{IKI}}, CV_{\text{IKI}}$
Quarantine Table
Blacklist: Suppress h_{rogue}
Anomaly ($\kappa \geq 8$)
Lockdown
Decision Boundary
$\mu < 25\,\text{ms} \vee CV < 0.15$
15s Watchdog Timer
Auto-Release on Idle
Rogue ($h \in \mathcal{Q}$)
[DROP EVENT]
Discard
Human ($h \notin \mathcal{Q}$)
[ALLOW EVENT]
Allow
Active Window
Foreground Target